



Do Audio-Visual Segmentation Models Truly Segment Sounding Objects?

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1. Introduction: The Problem with AVS

What is Audio Visual Segmentation

A multimodal task where a model must identify and *segment sounding objects* within a visual scene, mimicking the human ability to locate a sound's source.

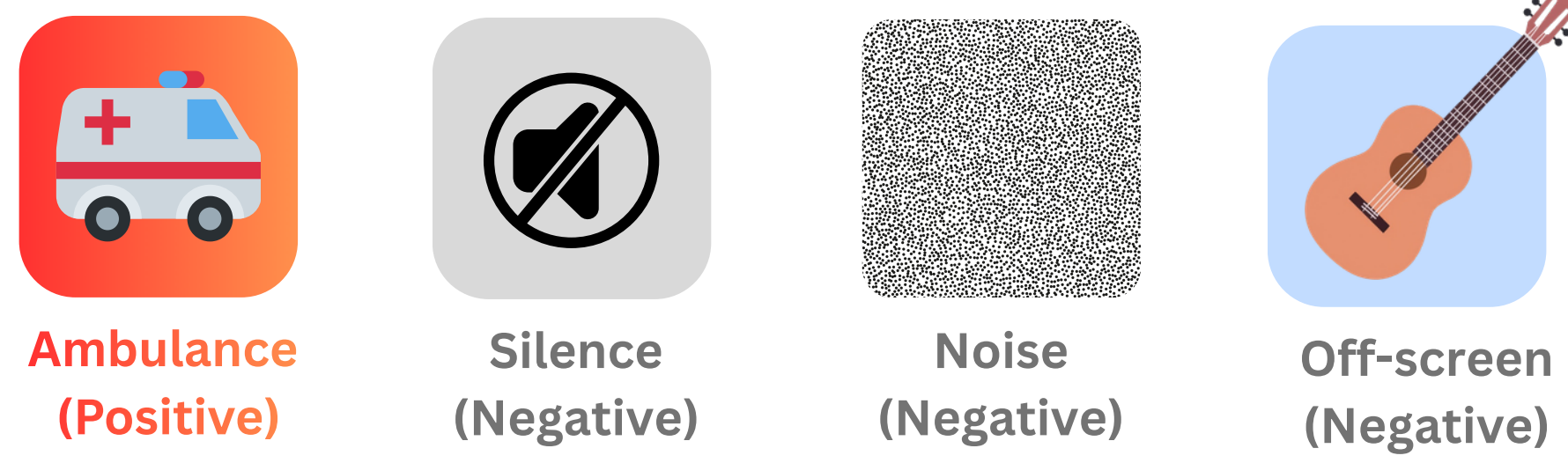
Current Model Exhibit a Strong “Visual Bias”

State-of-the-art (SOTA) models perform well on existing benchmarks but fail to genuinely integrate audio. They learn to segment visually salient objects, ignoring the audio context.

2. Benchmark: AVSBench-Robust

Four Distinct Audio Conditions

Evaluate models under both *standard* conditions and challenging *negative* audio scenarios.



New Global Metrics for Holistic Evaluation

False Positive Rate (FPR):

$$FPR = \frac{\sum_{x \in \mathcal{M}_{pred}} m(x)}{H \cdot W}, \quad (1)$$

Global mIoU (G-mIoU):

$$G-mIoU = \frac{2 \cdot mIoU_{\mathcal{P}} \cdot (1 - mIoU_{\mathcal{N}})}{mIoU_{\mathcal{P}} + (1 - mIoU_{\mathcal{N}})}. \quad (2)$$

Global F-score (G-F):

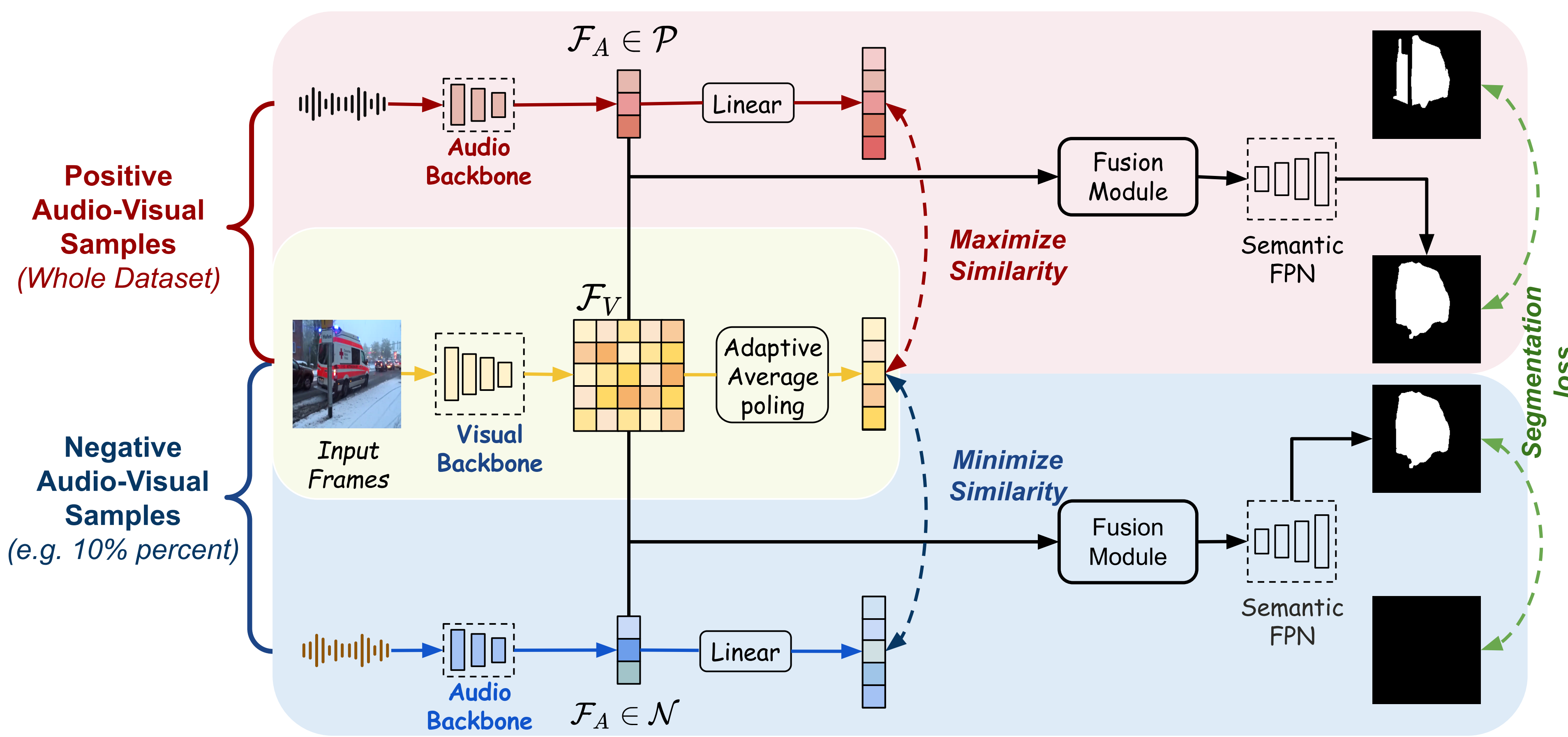
$$G-F = \frac{2 \cdot F_{\mathcal{P}} \cdot (1 - F_{\mathcal{N}})}{F_{\mathcal{P}} + (1 - F_{\mathcal{N}})}. \quad (3)$$

Global False Positive Rate (G-FPR):

$$G-FPR = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} FPR_i. \quad (4)$$

3. Model Arch: Our Method

A Framework for Learning True Audio-Visual Correspondence: Balanced Training with 10% Negative Samples



Ablation & Visualization

(1) Effects of negative samples and classifier guidance

	Negative samples	$\mathcal{L}_{BCE}$	G-mIoU $\uparrow$	G-F $\uparrow$	G-FPR $\downarrow$
S4	×	×	35.032	21.479	0.186
	✓	×	34.847	21.993	0.189
	✓	✓	87.672	82.461	0.000
MS3	×	×	59.468	51.036	0.072
	✓	×	55.489	30.057	0.095
	✓	✓	66.605	70.590	0.004

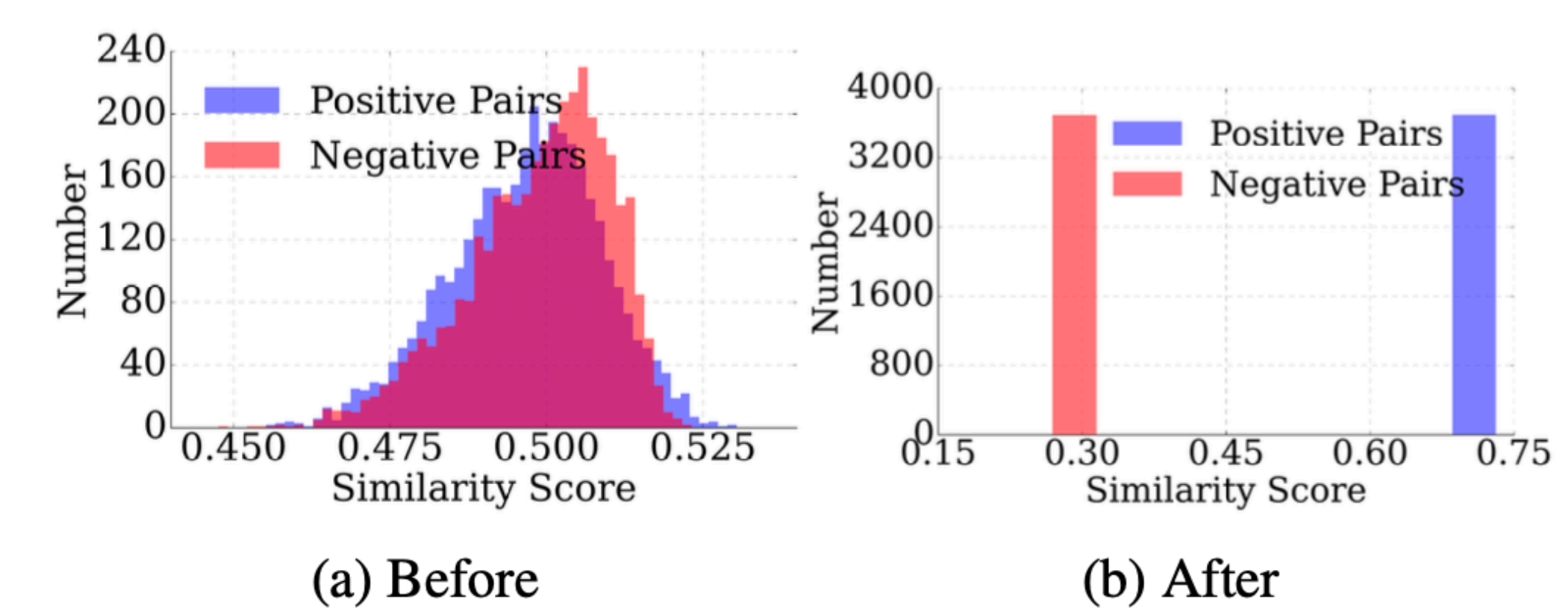
(2) Impact of the positive-negative pair ratio on model.

	Model	Pos pairs (%)	Neg pairs (%)	G-mIoU $\uparrow$	G-F $\uparrow$	G-FPR $\downarrow$
S4	Baseline	100	0	35.032	21.479	0.186
	Ours	90	10	87.672	82.461	0.000
		80	20	87.780	82.114	0.000
		70	30	87.204	82.233	0.000
MS3	Baseline	100	0	59.468	51.036	0.072
	Ours	90	10	65.427	70.911	0.001
		80	20	67.909	72.572	0.003
		70	30	66.251	72.908	0.000

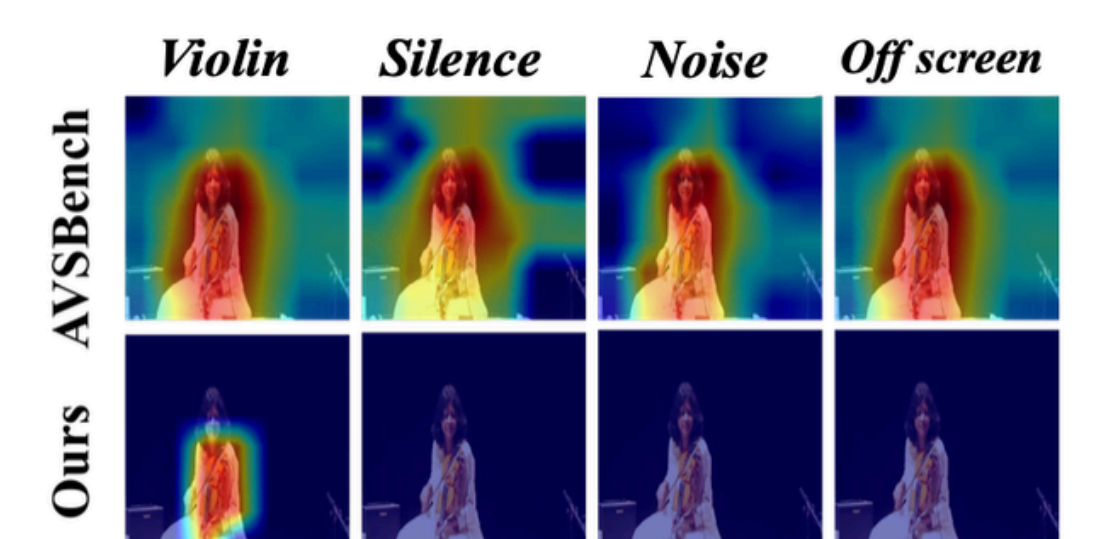
(3) Verifying True Correspondence vs. Pattern Memorization.

Audio Input Type	S4 Dataset mIoU	S4 Dataset F	MS3 Dataset mIoU	MS3 Dataset F
Original audio	78.15	88.22	51.27	64.51
Fused audio (original + noise)	77.18	86.44	50.76	63.60

(4) Learning Separable Feature Representations



(5) Qualitative comparison of audio-visual attention maps



Summary

- Uncovered Critical Visual Bias in SOTA AVS Models
- Proposed an Effective Solution for Robust AVS
- A Foundation for More Reliable Multimodal Systems

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4. Result in Table

Test set	Model	Global metric			Positive audio input		Unseen Audio		
		G-mIoU $\uparrow$	G-F $\uparrow$	G-FPR $\downarrow$	mIoU $\uparrow$	F-score $\uparrow$	mIoU $\downarrow$	F-score $\downarrow$	FPR $\downarrow$
AVSBench-S4	TPAVI (Zhou et al. 2022)	35.032	21.479	0.186	78.7	87.9	78.22	87.52	18.69
	AVSegFormer (Gao et al. 2024)	28.199	17.355	0.188	82.1	89.9	82.98	90.40	18.81
	Stepping-Stones (Ma et al. 2024)	28.980	15.806	0.190	83.2	91.3	82.51	91.27	19.02
	SAMA-AVS (Liu et al. 2024)	52.688	40.417	0.155	83.1	90.0	69.7	79.9	17.00
	CAVP (Chen et al. 2024)	33.526	19.891	0.185	78.7	88.8	78.7	88.8	18.53
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	COMBO (Yang et al. 2024)	26.062	14.888	0.190	84.7	91.9	84.6	91.9	18.9
	Selm (Li et al. 2024a)	41.543	30.054	0.175	76.6	86.2	71.5	81.8	17.67
	VCT (Huang et al. 2025)	23.791	12.329	0.193	86.2	93.4	86.2	93.4	19.71
	TPAVI + Ours	87.672	82.461	0.000	78.1	88.2	0.61	17.31	0.00
AVSBench-MS3	AVSegFormer + Ours	85.069	80.849	0.001	74.2	84.8	0.16	22.59	0.00
	TPAVI (Zhou et al. 2022)	59.468	51.036	0.072	54.0	64.5	42.24	62.16	9.53
	AVSegFormer (Gao et al. 2024)	54.889	46.571	0.103	61.3	73.8	51.14	66.24	11.73
	Stepping-Stones (Ma et al. 2024)	61.439	43.937	0.114	67.3	77.6	45.81	68.04	12.48
	SAMA-AVS (Liu et al. 2024)	65.308	65.038	0.125	68.6	78.3	41.9	47.5	14.36
	CAVP (Chen et al. 2024)	49.647	47.262	0.110	45.8	61.7	45.81	61.72	11.02
	Selm (Li et al. 2024a)	51.281	44.021	0.127	60.5	71.5	55.0	68.2	13.56
	VCT (Huang et al. 2025)	47.908	33.387	0.131	67.6	81.4	62.9	79.0	14.03
	TPAVI + Ours	65.427	70.911	0.001	51.3	64.5	9.06	33.96	0.82
	AVSegFormer + Ours	73.354	78.244	0.000	61.5	74.0	9.06	16.96	0.00